Media Capture in Local News Markets

Francesco Billari*

March 15, 2026

Please click here for the latest version.

Abstract

Online platform entry into advertising markets has drained revenues from traditional local newspapers, which remain key providers of local political information. This paper studies how declines in per-reader advertising revenues affect media bias when news outlets may be subject to political capture by incumbent politicians. I show that the relationship between advertising revenues and pro-incumbent media bias depends on both political rents and uncertainty about the quality of incumbent policy. When uncertainty is low and political rents are high, higher advertising revenues reduce media bias, consistent with standard theories of media capture. Otherwise, the relationship is non-monotonic and takes an inverted-U shape, reflecting the interaction between readership incentives and the compensation required to sustain distortion.

Keywords: media capture; Bayesian persuasion; political economy; local newspapers; advertising; subscriptions.

JEL Codes: D72, D83, L82, P16.

*Department of Economics, Cornell University. Email: fb293@cornell.edu.

1 Introduction

The digitalization of advertising markets has profoundly reshaped the economics of local journalism. The expansion of online platforms has diverted a large share of local advertising revenues away from traditional newspapers, weakening a key source of income for local news outlets. At the same time, local newspapers remain central providers of political information about municipal and regional governments. These developments raise a natural question: how do changes in newspapers' revenue structure affect the incentives to provide politically unbiased information?

This paper studies how per-reader advertising revenues influence media bias in local news markets where outlets may be subject to political capture by incumbent politicians. The analysis is motivated by the decline in local advertising revenues following the entry of online platforms such as Google and Facebook into local advertising markets. While these platforms have expanded the supply of information, they have also substantially reduced advertising revenues for traditional local outlets. Understanding how these revenue shocks affect the incentives of newspapers to provide political information is therefore crucial for assessing their implications for democratic accountability.

Most of the political economy literature on media capture treats advertising revenues as exogenous and focuses on how advertising dependence shapes editorial incentives. In contrast, I model the outlet's revenue structure as an endogenous choice between advertising and subscription income. This choice affects the equilibrium subscription price and, in turn, the median voter's readership decision and the amount of information available at the voting stage. As a result, the outlet's revenue structure determines the equilibrium level of political bias.

I develop a Bayesian persuasion model in which the quality of the incumbent's policy is uncertain and the newspaper chooses the informativeness of its reporting technology to maximize profits. The incumbent may offer transfers to the newspaper in exchange for more favorable coverage, while the median voter decides whether to read the newspaper and then votes according to the signal she observes. Advertising revenues affect the newspaper's pricing decisions and therefore the probability that the median voter reads the newspaper.

The analysis delivers two main results. First, when uncertainty about incumbent policy quality is sufficiently low – so that the informational value of the newspaper is limited – and the incumbent's office rent is sufficiently high, the relationship between advertising revenues and pro-incumbent media bias is negative. In this region, higher advertising revenues increase the opportunity cost of distorting coverage and therefore reduce equilibrium capture, consistent with the standard prediction in the political economy literature on media capture.

Second, outside this region – that is, whenever uncertainty about incumbent policy quality is sufficiently high or the incumbent's office rent is limited – the relationship between advertising revenues and media bias becomes non-monotonic and takes a reversed U-shape. The central idea behind this result is that advertising revenues act as a readership subsidy: higher advertising revenues favors newspaper circulation lowering the subscription price. Since the median voter trades off price against reporting quality, a decline in both reporting quality – through media

capture – and subscription price may therefore leave readership unchanged. However, distortion reduces editorial profits, and the media owner must be compensated by the incumbent for this loss. Importantly, the higher are advertising revenues, the greater is the compensation required to sustain a given level of distortion.

When advertising revenues are moderate, the electoral benefit from more favorable coverage outweighs the additional compensation cost, so the incumbent prefers lower reporting quality. When advertising revenues are sufficiently large, the compensation cost dominates the electoral gain, and the incumbent prefers less bias.

These results provide a new perspective on how shocks to advertising markets affect political information. In particular, the model implies that declines in advertising revenues may lead newspapers to increase subscription prices and provide more informative coverage in order to preserve readership. This mechanism is consistent with empirical evidence documenting that the entry of online platforms into local advertising markets is associated with a rise in subscription prices and, most importantly, a decline in local political news coverage. Within the framework of the model, such a decline may be interpreted as a reduction in pro-incumbent bias.

Literature contribution This paper contributes to several strands of the literature on media economics and political economy.

First, it relates to work studying how shocks to advertising markets affect newspapers' business strategies and editorial content. Seamans and Zhu (2014) document that the entry of Craigslist increased subscription prices of U.S. newspapers following the collapse of classified advertising revenues. Similarly, Djourelova, Durante, and Martin (2023) show that the same shock led newspapers to reduce the production of political news coverage. My analysis bridges these two findings by showing how newspapers' economic responses to changes in advertising markets affect the informational content of news, and provides an alternative interpretation of the decline in political news coverage.

Second, the paper speaks to the broader literature on the relationship between advertising revenues and media independence. Historically, advertising has been viewed as a force promoting editorial independence. For example, Hamilton (2004) argues that the expansion of advertising markets in the late nineteenth century played a key role in the emergence of an independent commercial press. Similarly, Petrova (2011) documents that the growth of advertising revenues contributed to the rise of politically independent newspapers. Using evidence from France, Angelucci and Cagé (2019) show that the introduction of television advertising reduced newspapers' production of journalistic-intensive content. By studying the interaction between advertising markets and the incentives of media producers, this paper contributes to the debate on whether advertising revenues ultimately enhance or distort the informational role of the media.

Third, this paper contributes to the theoretical literature on media capture and political influence over news production. In a seminal contribution, Besley and Prat (2006) develop a model in which governments can bribe media outlets to suppress unfavorable news, showing that larger advertising markets can reduce media capture by raising the opportunity cost of bias. My ap-

proach is closely related to Gehlbach and Sonin (2014), who analyze government influence over media content when outlets rely on advertising revenues and allow for changes in media ownership (government versus private). They show that the relationship between advertising revenues and media bias is not monotonic, although conditional on ownership status the relationship is negative. Unlike their framework, the media owner in my model is always private and the newspaper's revenue structure is endogenous, so that advertising is not the only source of income. In contrast to their finding, I show that the relationship between advertising revenues and media bias can be positive. Petrova (2012) derives conditions under which the relationship between advertising revenues and media bias is positive or negative. In environments like the one studied here – where audience size and bias are substitutes – her model predicts a negative relationship. In contrast, I derive conditions under which the relationship between advertising revenues and media bias can be negative or non-monotonic, potentially taking a reversed U-shape.

Finally, the paper relates to empirical work documenting media capture in democratic settings. Di Tella and Franceschelli (2011) show that government advertising in Argentina is negatively correlated with newspaper coverage of corruption scandals, suggesting that financial transfers can distort news content. More recently, Szeidl and Szucs (2021) provide evidence from Hungary that governments allocate advertising to politically connected outlets which in turn provide favorable coverage, consistent with a mechanism of favor exchange between governments and media firms.

The remainder of the paper is organized as follows. Section 2 introduces the model. Section 3 studies the benchmark case in which the newspaper relies exclusively on advertising revenues. Section 4 examines the case of subscription-based revenues. Section 5 analyzes the general case with mixed revenues from advertising and subscriptions.

2 Model

Environment. The economy consists of three players: a median voter, an incumbent politician, and a media owner. The incumbent's policy quality is determined by a state of the world $\vartheta \in \{0, 1\}$. If $\vartheta = 1$, the incumbent's policy is good; if $\vartheta = 0$, it is bad. The state is unknown to all players. The common prior belief is $\mathbb{P}(\vartheta = 1) = \pi \in (0, 1/2)$. The restriction $\pi < 1/2$ ensures that, in the absence of additional information, the median voter would not re-elect the incumbent, consistent with the approach in Gehlbach and Sonin (2014).[1] If re-elected, the incumbent obtains an office rent $R > 0$. Otherwise, her payoff is zero.

[1] In many political environments prior beliefs may instead favor the incumbent or the candidate in office, so that $\pi \geq 1/2$. One way to reconcile this possibility with the baseline assumption is to interpret the model as one in which the incumbent wins only if the median voter turns out and votes for him. Suppose voting involves a cost that is sufficiently high that the prior belief alone does not justify turning out, even when $\pi \geq 1/2$. In that case the incumbent needs the voter to acquire information – by reading the newspaper – in order to induce turnout. For completeness, Appendix B analyzes the case in which voting costs are zero and $\pi \geq 1/2$, showing that the main qualitative results of the model remain unchanged.

Information Structure and Reporting Technology. The newspaper produces a binary signal $s \in \{0, 1\}$ about the state ϑ. Reporting quality is denoted by $q \in [0, 1]$ and determines the precision of the signal when the incumbent's policy is bad. Specifically, the signal structure is given by

$$\mathbb{P}(s = 1 \mid \vartheta = 1) = 1,$$

$$\mathbb{P}(s = 0 \mid \vartheta = 0) = q.$$

Thus, if the incumbent's policy is good, the signal is perfectly informative. If the policy is bad, the signal correctly reports this with probability q and instead sends a favorable signal with probability $1 - q$. Lower values of q therefore correspond to greater pro-incumbent distortion and can be interpreted as a higher degree of media capture. Let μ_s denote the posterior belief that $\vartheta = 1$ after observing signal s.

Voter Preferences and Information Demand. The median voter chooses whether to read the newspaper and whether to re-elect the incumbent. Voting preferences are represented by quadratic loss: $u(v, \vartheta) = -(v - \vartheta)^2$, where $v \in \{0, 1\}$ denotes the voting decision, with $v = 1$ corresponding to re-election.

If the voter reads the newspaper, she incurs an attention cost c drawn from a uniform distribution, $c \sim U\left[-\frac{1}{2}, \frac{1}{2}\right]$. Positive realizations correspond to the conventional cost of acquiring information, whereas negative realizations capture intrinsic demand for news consumption, reflecting habit formation or civic interest. The symmetric support ensures an interior readership margin.

After observing s, the voter chooses the optimal voting action. Standard arguments imply that the voter re-elects the incumbent if and only if $\mu_s \geq 1/2$.

Lemma 1. *If the voter reads and observes $s = 1$, she re-elects the incumbent if and only if $q \geq \underline{q}(\pi)$, where $\underline{q}(\pi) := \frac{1-2\pi}{1-\pi}$. Moreover, for $q < \underline{q}(\pi)$, lowering reporting quality does not increase the incumbent's probability of re-election.*

Proof. See Appendix A. □

Let $\tilde{c}(\pi, q)$ denote the cutoff attention cost that makes the voter indifferent between reading and not reading. The voter reads if and only if $c \leq \tilde{c}(\pi, q)$. This cutoff depends on the informational value of the signal, which is itself determined by the prior π and reporting quality q.

Media Revenues. The newspaper can generate revenues from subscription fees and advertising. Let $p \geq 0$ denote the subscription price charged to readers and let $\alpha \geq 0$ denote advertising revenue per reader. If the voter reads the newspaper, the media owner earns $p + \alpha$. Given (p, q, α), expected editorial profits are $\Pi(p, q; \alpha) = (p + \alpha)\mathbb{P}\big(c \leq \tilde{c}(\pi, q) - p\big)$. The probability term reflects that the voter reads only if the net benefit of information exceeds the subscription price. Different revenue regimes correspond to restrictions on (p, α). The advertising-only benchmark sets $p = 0$,

the subscription-only benchmark sets $\alpha = 0$, and the mixed-revenue case allows both components to be active.

Political Contracting and Media Capture. The incumbent can attempt to influence reporting quality through transfers. She offers a contract $(\tilde{q}, T) \in [0,1] \times \mathbb{R}_+$ consisting of a target reporting quality $\tilde{q}$ and a transfer T. The media owner chooses reporting quality q after observing the offer.

The media owner accepts the contract if and only if the transfer compensates for the loss in editorial profits relative to the optimal unconstrained choice of q. Formally, the contract is accepted whenever $T \geq \Pi(p, q; \alpha) - \Pi(p, \tilde{q}; \alpha)$. Lower reporting quality increases the incumbent's probability of re-election but may reduce readership and therefore editorial revenues.

Timing and Equilibrium Concept. The timing of the game is as follows:

1. The newspaper sets the subscription price $p \geq 0$.
2. The incumbent offers a contract $(\tilde{q}, T) \in [0,1] \times \mathbb{R}_+$.
3. The media owner chooses reporting quality $q \in [0,1]$.
4. The attention cost c is realized and the voter decides whether to read.
5. If the voter reads, the signal s is realized.
6. The voter chooses whether to re-elect the incumbent.

The assumption that the subscription price is set at the outset of the game ensures that pricing decisions are not part of the political contracting stage between the incumbent and the media owner. In particular, the incumbent can influence the newspaper only through transfers that affect the informativeness of reporting, but not through the outlet's pricing policy. This restriction isolates the mechanism of interest – political incentives to distort information – while keeping pricing decisions independent of political influence. The assumption is consistent with the existing media capture literature, which typically treats pricing decisions as outside the scope of political bargaining (see, e.g., Petrova (2012)).

3 Advertising-Only Benchmark

In this section I characterize equilibrium when the newspaper relies exclusively on advertising revenues. This corresponds to the restriction $p = 0$, so that editorial profits depend only on advertising revenue per reader $\alpha \geq 0$. The purpose of this benchmark is to isolate how capture operates when readership is effectively subsidized by advertising.

Revenue Restriction and Timing. Setting $p = 0$, the newspaper earns α for each reader. Expected editorial profits are therefore $\Pi(q;\alpha) = \alpha\mathbb{P}\big(c \leq \tilde{c}(\pi, q)\big)$.

The timing simplifies as follows:

1. The incumbent offers a contract $(\tilde{q}, T)$.
2. The media owner chooses reporting quality q.
3. The attention cost c is realized and the voter decides whether to read.
4. If the voter reads, the signal s is realized.
5. The voter chooses whether to re-elect the incumbent.

Voter Behavior and Electoral Probability. Given reporting quality q, the voter updates beliefs upon observing the signal and re-elects the incumbent whenever the posterior satisfies $\mu_s \geq 1/2$. By Lemma 1, the voter re-elects the incumbent after $s = 1$ if and only if $q \geq \underline{q}(\pi)$.

Let $\tilde{c}(\pi, q)$ denote the cutoff attention cost such that the voter is indifferent between reading and not reading. Since $c \sim U[-1/2, 1/2]$, readership is given by $\mathbb{P}(c \leq \tilde{c}(\pi, q)) = \tilde{c}(\pi, q) + \frac{1}{2}$. Let $\phi(\pi, q)$ denote the probability that the incumbent is re-elected. This probability depends on both the informativeness of the signal and the fraction of voters who choose to read. In particular, $\phi(\pi, q) = \big[\pi + (1-\pi)(1-q)\big]\mathbb{P}(c \leq \tilde{c}(\pi, q))$.

Media Owner's Problem. For a given contract $(\tilde{q}, T)$, the media owner chooses reporting quality to maximize $T \cdot \mathbb{1}\{q = \tilde{q}\} + \Pi(q;\alpha)$. The contract is accepted if and only if the transfer compensates the loss in advertising revenues induced by the deviation from the unconstrained optimal quality. The minimum transfer required to implement $\tilde{q}$ is therefore $T \geq \Pi(q^*;\alpha) - \Pi(\tilde{q};\alpha)$, where q^* denotes the profit-maximizing reporting quality absent political influence.

Incumbent's Problem. Anticipating the required transfer, the incumbent chooses $\tilde{q}$ to maximize $R\phi(\pi, \tilde{q}) - T(\tilde{q})$, where $T(\tilde{q})$ is the minimum transfer necessary to induce reporting quality $\tilde{q}$. The incumbent trades off the electoral benefit from lowering reporting quality against the higher compensation required to offset reduced advertising revenues.

Proposition 1. *Assume revenues are advertising-based $p = 0$, and define $q^* = \frac{\alpha - R(2\pi - 3/2)}{2R(1-\pi)}$. Then equilibrium reporting quality is given by*

$$q^A = \begin{cases} 1 & \text{if } R \leq 2\alpha, \\ \underline{q}(\pi) & \text{if } \pi < 1/4 \text{ and } R \geq \frac{2\alpha}{1-4\pi}, \\ q^* & \text{otherwise.} \end{cases}$$

Proof. See Appendix A. □

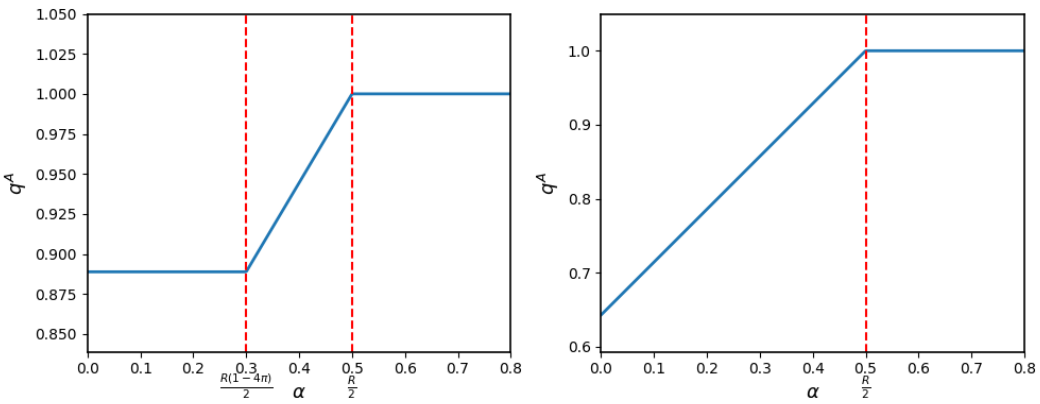


Figure 1: Equilibrium reporting quality with advertising-based revenues. The figure plots q^A as a function of advertising revenues α. In the left panel ($\pi < 1/4$), equilibrium reporting quality is equal to $\underline{q}(\pi)$ for low levels of advertising revenues, then follows the interior solution q^*, and finally reaches the upper bound 1 once $\alpha \geq R/2$. In the right panel ($\pi > 1/4$), equilibrium reporting quality follows q^* for low advertising revenues and reaches 1 once $\alpha \geq R/2$.

Corollary 1. *Assume revenues are advertising-based ($p = 0$) and fix (R, π). Equilibrium reporting quality $q^A(\alpha)$ is weakly increasing in advertising revenues α. Equivalently, equilibrium media bias $1 - q^A(\alpha)$ is weakly decreasing in α.*

Proof. Fix (R, π). Rewriting the parameter restrictions in Proposition 1 in terms of α yields two thresholds. First, $R \geq \frac{2\alpha}{1-4\pi}$ is equivalent to $\alpha \leq \frac{R(1-4\pi)}{2}$. Second, $R \leq 2\alpha$ is equivalent to $\alpha \geq \frac{R}{2}$. When $\pi \in (0, 1/4)$, reporting quality is constant at its lower bound for $\alpha \leq \frac{R(1-4\pi)}{2}$, strictly increasing in α on the intermediate region, and equal to 1 for $\alpha \geq \frac{R}{2}$. When $\pi \in [1/4, 1/2)$, reporting quality is strictly increasing in α for $\alpha < \frac{R}{2}$ and equal to 1 thereafter. In both cases, $q^A(\alpha)$ is weakly increasing in α. □

Higher advertising revenues increase the marginal value of readership. Since lower reporting quality reduces readership, the transfer required to sustain capture increases with α. For a given political rent R, equilibrium distortion is therefore reduced as advertising revenues rise, and eventually capture cannot be sustained once α exceeds $\overline{\alpha}(R)$.

4 Subscription-Only Benchmark

In this section I characterize equilibrium when the newspaper relies exclusively on subscription revenues, i.e. $\alpha = 0$. Unlike the advertising benchmark, the newspaper now earns revenue only from readers and therefore sets the subscription price at the outset of the game. The purpose of this benchmark is to isolate how capture operates when readership directly determines revenue rather than being subsidized by advertising.

Revenue Restriction and Timing. Setting $\alpha = 0$, expected editorial profits are $\Pi(p, q) = p\mathbb{P}\big(c \leq \tilde{c}(\pi, q) - p\big)$.

The timing differs slightly from the advertising benchmark:

1. The newspaper sets the subscription price $p \geq 0$.
2. The incumbent offers a contract $(\tilde{q}, T)$.
3. The media owner chooses reporting quality q.
4. The attention cost c is realized and the voter decides whether to read.
5. If the voter reads, the signal s is realized.
6. The voter chooses whether to re-elect the incumbent.

Voter Behavior and Electoral Probability. The voter's updating and persuasion constraint are identical to the advertising benchmark. In particular, the voter re-elects the incumbent if and only if $s = 1$ and $q \geq \frac{1-2\pi}{1-\pi}$.

Given (p, q), the voter reads if and only if $c \leq \tilde{c}(\pi, q) - p$. Since $c \sim U[-1/2, 1/2]$, readership equals $\mathbb{P}(c \leq \tilde{c}(\pi, q) - p) = \frac{1}{2} + \tilde{c}(\pi, q) - p$. Let $\phi(\pi, q)$ denote the probability of re-election. As in the advertising benchmark, $\phi(\pi, q) = \big[\pi + (1 - \pi)(1 - q)\big]\mathbb{P}(c \leq \tilde{c}(\pi, q) - p)$.

Media Owner's Problem. For a given (p, q), subscription profits are $\Pi(p, q) = p\Big(\frac{1}{2} + \tilde{c}(\pi, q) - p\Big)$. For fixed reporting quality q, the profit-maximizing price solves

$$\frac{\partial \Pi}{\partial p} = \frac{1}{2} + \tilde{c}(\pi, q) - 2p = 0,$$

yielding $p^*(q) = \frac{1}{2}\Big(\frac{1}{2} + \tilde{c}(\pi, q)\Big)$. Substituting $\tilde{c}(\pi, q) = (1 - \pi)q - (1 - 2\pi)$ gives $p^*(q) = \frac{1}{4} + \frac{1}{2}\big((1 - \pi)q - (1 - 2\pi)\big)$. Absent political influence, the media owner chooses $q = 1$ and sets price accordingly. The minimum transfer required to induce reporting quality $\tilde{q}$ equals the foregone subscription profit relative to $q = 1$.

Incumbent's Problem. Anticipating optimal pricing, the incumbent chooses $\tilde{q}$ to maximize $R\phi(\pi, \tilde{q}) - T(\tilde{q})$, where $T(\tilde{q})$ equals the loss in subscription profits from deviating from $q = 1$. Relative to the advertising benchmark, subscription pricing internalizes readership more directly, altering the capture trade-off.

Proposition 2. *Assume revenues are subscription-based $\alpha = 0$, and let $q^{S,*} = \frac{R(7/4 - 3\pi/2) + 1/4 + \pi/2}{2R(1-\pi)}$.*

Then equilibrium reporting quality is given by

$$q^S = \begin{cases} 1 & \text{if } R \leq \dfrac{1+2\pi}{1-2\pi}, \\ \underline{q}(\pi) & \text{if } \pi < 1/10 \text{ and } R \geq \dfrac{1+2\pi}{1-10\pi}, \\ q^{S,*} & \text{otherwise.} \end{cases}$$

Proof. See Appendix A. □

When $\pi < 1/10$, the ex ante informational value of the newspaper is low, which constrains the subscription price and compresses subscription profits. As a consequence, the opportunity cost of distortion is small: reducing reporting quality lowers readership and profits only marginally. If political rents are sufficiently high, the incumbent finds it optimal to purchase maximal feasible distortion, leading to the corner solution $q^S = \underline{q}(\pi)$.

The proof follows the same steps as in Proposition 1 and is reported in Appendix A.

5 Mixed Revenues

In this section I characterize equilibrium when the newspaper can generate revenues from both subscriptions and advertising. The purpose of this section is to study how the composition of revenues affects capture incentives when the subscription price is chosen at the outset and advertising revenues act as a readership subsidy.

Revenue Restriction and Timing. Let $\alpha \geq 0$ denote advertising revenue per reader and let $p \geq 0$ denote the subscription price. If the voter reads, the media owner earns $p + \alpha$. Expected editorial profits are therefore $\Pi(p, q; \alpha) = (p + \alpha)\mathbb{P}\big(c \leq \tilde{c}(\pi, q) - p\big)$.

The timing is the same as in the subscription benchmark:

1. The newspaper sets the subscription price $p \geq 0$.
2. The incumbent offers a contract $(\tilde{q}, T)$.
3. The media owner chooses reporting quality q.
4. The attention cost c is realized and the voter decides whether to read.
5. If the voter reads, the signal s is realized.
6. The voter chooses whether to re-elect the incumbent.

Voter Behavior and Electoral Probability. Voter updating and the persuasion constraint are unchanged. In particular, by Lemma 1, it suffices to restrict attention to $q \in [\underline{q}(\pi), 1]$, where $\underline{q}(\pi) = \frac{1-2\pi}{1-\pi}$. Given (p, q), the voter reads if and only if $c \leq \tilde{c}(\pi, q) - p$, so readership is $\mathbb{P}(c \leq \tilde{c}(\pi, q) - p)$. The probability of re-election is

$$\phi(\pi, q) = \big[\pi + (1-\pi)(1-q)\big]\mathbb{P}\big(c \leq \tilde{c}(\pi, q) - p\big).$$

Pricing. For any reporting quality q, the newspaper chooses p at the outset to maximize $\Pi(p, q; \alpha)$. Let $\Delta := \frac{1}{2} - \pi$. Because $\pi = \frac{1}{2}$ represents maximal uncertainty about incumbent policy quality, Δ measures the distance from maximal uncertainty. In particular, Δ can be interpreted as a proxy for the ex ante informational value of an unbiased signal: larger values of Δ correspond to environments in which uncertainty is lower and the informational value of news is reduced.The revenue-maximizing price is

$$p^* = \max\left\{0, \frac{1 - \Delta - \alpha}{2}\right\}.$$

This implies that sufficiently large advertising revenues reduce the optimal subscription price, and for $\alpha \geq 1 - \Delta$ the newspaper optimally sets $p^* = 0$.

Political Contracting and Equilibrium. Given the price p^*, the capture stage proceeds as in the previous benchmarks. The incumbent trades off higher re-election probability from lower reporting quality against the transfer required to compensate the media owner for foregone editorial profits. The following proposition characterizes equilibrium reporting quality.

Proposition 3. *Assume endogenous revenue composition. Let* $\Delta := \frac{1}{2} - \pi$ *and define*

$$\underline{R}(\alpha, \pi) := \frac{1 + 2\alpha + 2\pi}{1 + 2\alpha - 2\pi}, \qquad \overline{R}(\alpha, \pi) := \frac{1 + 2\alpha + 2\pi}{1 + 2\alpha - 10\pi},$$

and

$$q^{M,*}(\alpha, \pi) := \frac{R(7 - 6\pi - 2\alpha) + (1 + 2\alpha + 2\pi)}{8R(1 - \pi)}.$$

If $\alpha \geq 1 - \Delta$*, then* $p^* = 0$ *and equilibrium reporting quality coincides with the advertising-only benchmark, i.e.* $q^M = q^A$*. Suppose instead that* $\alpha < 1 - \Delta$*.*

(i) If $\pi < \frac{1}{10} + \frac{\alpha}{5}$*, equilibrium reporting quality is*

$$q^M = \begin{cases} 1 & \text{if } R \leq \underline{R}(\alpha, \pi), \\ q^{M,*}(\alpha, \pi) & \text{if } \underline{R}(\alpha, \pi) < R < \overline{R}(\alpha, \pi), \\ \underline{q}(\pi) & \text{if } R \geq \overline{R}(\alpha, \pi). \end{cases}$$

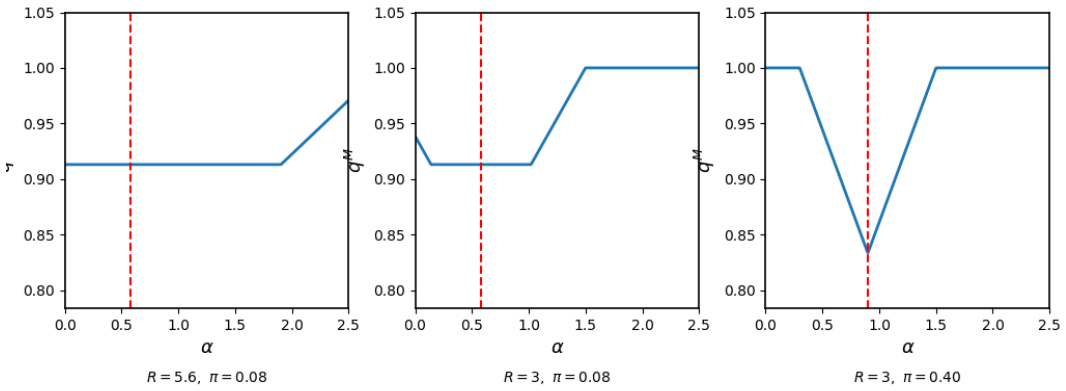


Figure 2: Reporting quality as a function of advertising revenues. The left panel shows a weakly monotonic increase in reporting quality as advertising revenues rise. In contrast, the middle and right panels display a U-shaped relationship between advertising revenues and reporting quality. The vertical red dashed line marks the threshold level of revenues beyond which the outlet's revenue structure depends exclusively on advertising.

(ii) If $\pi \geq \frac{1}{10} + \frac{\alpha}{5}$*, equilibrium reporting quality is*

$$q^M = \begin{cases} 1 & \text{if } R \leq \underline{R}(\alpha, \pi), \\ q^{M,*}(\alpha, \pi) & \text{if } R > \underline{R}(\alpha, \pi). \end{cases}$$

Proof. See Appendix A. □

Proposition 4. *Assume endogenous revenue composition. (i) If* $R \leq 1 + 2\pi$*, then capture never occurs and* $q^M(\alpha) = 1$ *for every* $\alpha \geq 0$*. (ii) If* $\pi < 1/10$ *and* $R \geq \frac{1+2\pi}{1-10\pi}$*, then* $q^M(\alpha)$ *is weakly increasing in* α*. (iii) Otherwise, there exists* $\tilde{\alpha}(R)$ *such that* $q^M(\alpha)$ *is decreasing in* α *for* $\alpha \leq \tilde{\alpha}(R)$ *and weakly increasing for* $\alpha > \tilde{\alpha}(R)$*.*

Proof. See Appendix A. □

The relationship between advertising revenues and media bias depends critically on both the level of political rents and the degree of uncertainty about incumbent policy quality.

Advertising revenues act as a readership subsidy: higher advertising revenues allow the newspaper to lower the subscription price, and the median voter trades off price against reporting quality. A decline in both reporting quality and subscription price may therefore leave readership unchanged. However, distortion reduces editorial profits, and the media owner must be compensated by the incumbent for this loss. Importantly, the higher are advertising revenues, the greater is the compensation required to sustain a given level of distortion.

When advertising revenues are moderate, the electoral benefit from more favorable coverage outweighs the additional compensation cost, so the incumbent prefers lower reporting quality. When advertising revenues are sufficiently large, the compensation cost dominates the electoral

gain, and the incumbent prefers less bias. This generates the non-monotonic relationship described in Proposition 4.

The comparison with the subscription-only benchmark clarifies the role of uncertainty. When uncertainty about incumbent quality is low and political rents are sufficiently high, the incumbent optimally purchases maximal feasible distortion even in the absence of advertising revenues. In this region, additional advertising revenues do not increase the electoral benefit of distortion but only raise the compensation cost. Consequently, reporting quality is weakly increasing in advertising revenues.

6 Conclusion

This paper studies how the revenue structure of local newspapers affects the incentives to provide politically unbiased information. The analysis is motivated by the transformation of advertising markets following the entry of online platforms, which has substantially reduced advertising revenues for traditional local news outlets. Understanding how such revenue shocks affect information provision is important for assessing their implications for democratic accountability.

I develop a model in which a monopolistic newspaper chooses the informativeness of its reporting technology while facing the possibility of political capture by an incumbent politician. The key feature of the model is that the median voter's readership decision determines whether the incumbent can remain in office. As a result, the newspaper's revenue structure – specifically the balance between advertising and subscription revenues – affects subscription prices, readership incentives, and ultimately the equilibrium level of media bias.

The analysis shows that the relationship between advertising revenues and pro-incumbent media bias depends on both the incumbent's office rent and the level of uncertainty about policy quality. When uncertainty is low and political rents are sufficiently large, higher advertising revenues increase the opportunity cost of distorting information and therefore reduce equilibrium media bias. In this region, the model reproduces the standard prediction in the political economy literature on media capture. However, when uncertainty about policy quality is sufficiently high or political rents are limited, the relationship becomes non-monotonic and takes a reversed U-shape. Advertising revenues initially increase bias by subsidizing readership and reducing the cost of capture, but beyond a threshold further increases raise the compensation required to sustain distortion and ultimately reduce media bias.

These results provide a new perspective on how changes in the economics of local journalism affect political information. In particular, the model suggests that shocks that reduce advertising revenues – such as the entry of online platforms into local advertising markets – may induce newspapers to rely more heavily on subscription income and to provide higher-quality information in order to maintain readership. In this sense, declines in advertising revenues may lead to reductions in pro-incumbent bias, even if they are accompanied by reductions in the volume of political news coverage.

More broadly, the analysis highlights the importance of considering the interaction between media market structure, revenue models, and political incentives when evaluating the consequences of technological change in the media industry. Future research could extend the framework to settings with multiple competing outlets, heterogeneous voters, or strategic interactions between media firms and political actors. Such extensions may help further clarify the role of media institutions in shaping the quality of democratic information.

References

Angelucci, C. and J. Cagé (2019). "Newspapers in Times of Low Advertising Revenues." *American Economic Journal: Economic Policy*, 11(3), 319–364.

Besley, T. and A. Prat (2006). "Handcuffs for the Grabbing Hand? Media Capture and Government Accountability." *American Economic Review*, 96(3), 720–736.

Di Tella, R. and I. Franceschelli (2011). "Government Advertising and Media Coverage of Corruption Scandals." *American Economic Journal: Applied Economics*, 3(4), 119–151.

Djourelova, M., R. Durante, and G. Martin (2023). "The Impact of the Internet on Political News Coverage: Evidence from the Introduction of Craigslist." *American Economic Journal: Applied Economics*, 15(2), 1–41.

Gehlbach, S. and K. Sonin (2014). "Government Control of the Media." *Journal of Public Economics*, 118, 163–171.

Hamilton, J. T. (2004). *All the News That's Fit to Sell: How the Market Transforms Information into News*. Princeton University Press.

Petrova, M. (2011). "Newspapers and Parties: How Advertising Revenues Created an Independent Press." *American Political Science Review*, 105(4), 790–808.

Petrova, M. (2012). "Mass Media and Special Interest Groups." *Journal of Economic Behavior and Organization*, 84(1), 17–38.

Seamans, R. and F. Zhu (2014). "Responses to Entry in Multi-Sided Markets: The Impact of Craigslist on Local Newspapers." *Management Science*, 60(2), 476–493.

Szeidl, A. and F. Szucs (2021). "Media Capture Through Favor Exchange." *Econometrica*, 89(1), 281–310.

A Proofs

A.1 Proof of Lemma 1

Proof. If the voter reads and observes $s = 1$, Bayes' rule implies $\mu_1 = \frac{\pi}{\pi+(1-\pi)(1-q)}$. The voter re-elects the incumbent if and only if $\mu_1 \geq 1/2$, which holds if and only if $q \geq \frac{1-2\pi}{1-\pi} = \underline{q}(\pi)$. For $q < \underline{q}(\pi)$, even after observing $s = 1$ the voter chooses $v = 0$, so lowering q further does not increase the probability of re-election. □

A.2 Proof of Proposition 1

Proof. Assume revenues are advertising-based ($p = 0$). By Lemma 1, it suffices to restrict attention to $q \in [\underline{q}(\pi), 1]$.

Step 1. The cutoff attention cost is $\tilde{c}(\pi, q) = \pi-(1-\pi)(1-q) = (1-\pi)q-(1-2\pi)$, and readership equals $\Pr(c \leq \tilde{c}(\pi, q)) = (1-\pi)q + \left(2\pi - \frac{1}{2}\right)$.

Proof of Step 1. If the voter does not read, she votes according to the prior. Since $\pi < 1/2$, the optimal uninformed vote is $v = 0$, yielding expected payoff $\mathbb{E}[u(0, \vartheta)] = -\pi$. If the voter reads, she pays cost c and votes optimally. Under $q \geq \underline{q}(\pi)$, she chooses $v = 1$ after $s = 1$ and $v = 0$ after $s = 0$. The probability of $s = 1$ is $\Pr(s = 1) = \pi + (1-\pi)(1-q)$. Expected payoff from reading (before cost) is $\mathbb{E}[u \mid r = 1] = -\Pr(s = 1)\left(1 - \frac{\pi}{\Pr(s=1)}\right) = -(1-\pi)(1-q)$. Hence the net benefit from reading is $\pi-(1-\pi)(1-q)$, and the voter reads if and only if $c \leq \tilde{c}(\pi, q) = \pi-(1-\pi)(1-q)$. Since $c \sim U[-1/2, 1/2]$, $\Pr(c \leq \tilde{c}(\pi, q)) = \frac{1}{2} + \tilde{c}(\pi, q) = (1-\pi)q + \left(2\pi - \frac{1}{2}\right)$ ■

Step 2. Advertising profits are strictly increasing in q, and the minimum transfer required to induce reporting quality $\tilde{q}$ equals $T(\tilde{q}) = \alpha(1-\pi)(1-\tilde{q})$.

Proof of Step 2. Advertising profits are $\Pi(q; \alpha) = \alpha \Pr(c \leq \tilde{c}(\pi, q))$. Since readership is increasing in q, $\Pi(q; \alpha)$ is strictly increasing in q. Hence the media owner chooses $q = 1$ absent political influence. The minimum transfer is therefore $T(\tilde{q}) = \Pi(1; \alpha) - \Pi(\tilde{q}; \alpha) = \alpha(1-\pi)(1-\tilde{q})$. ■

Step 3. The incumbent's objective is strictly concave in q, and the unique interior candidate is

$$q^* = \frac{\alpha - R(2\pi - 3/2)}{2R(1-\pi)}.$$

Proof of Step 3. If $q \geq \underline{q}(\pi)$, the re-election probability equals $\phi(\pi, q) = \left[\pi + (1-\pi)(1-q)\right] \Pr(c \leq \tilde{c}(\pi, q))$. The incumbent maximizes $F(q) = R\phi(\pi, q) - \alpha(1-\pi)(1-q)$ over $[\underline{q}(\pi), 1]$. Differentiation yields $F'(q) = (1-\pi)\left[R\left(\frac{3}{2} - 2\pi - 2(1-\pi)q\right) + \alpha\right]$, which is strictly decreasing in q. Solving $F'(q) = 0$ gives $q^* = \frac{\alpha - R(2\pi-3/2)}{2R(1-\pi)}$. ■

Step 4. Equilibrium reporting quality is given by:

$$q^A = \begin{cases} 1 & \text{if } R \leq 2\alpha, \\ \underline{q}(\pi) & \text{if } \pi < 1/4 \text{ and } R \geq \frac{2\alpha}{1-4\pi}, \\ q^* & \text{otherwise.} \end{cases}$$

Proof of Step 4. Since F is strictly concave, the optimum is determined by the sign of $F'(q)$ at the boundaries. At $q = 1$, $F'(1) = (1-\pi)\left(\alpha - \frac{R}{2}\right)$. Hence $F'(1) \geq 0$ if and only if $R \leq 2\alpha$, implying $q^A = 1$. At $q = \underline{q}(\pi)$, $F'(\underline{q}(\pi)) = (1-\pi)\left[R\left(2\pi - \frac{1}{2}\right) + \alpha\right]$. Thus $F'(\underline{q}(\pi)) \leq 0$ if and only if $R \geq \frac{2\alpha}{1-4\pi}$. This condition can hold only if $\pi < 1/4$, implying $q^A = \underline{q}(\pi)$. Otherwise, the unique interior solution q^* applies. ■

Collecting the cases yields the characterization in Proposition 1. □

A.3 Proof of Proposition 2

Proof. Assume revenues are subscription-based ($\alpha = 0$). By Lemma 1, it suffices to restrict attention to $q \in [\underline{q}(\pi), 1]$.

Step 1. Given (p, q), the voter reads if and only if $c \leq \tilde{c}(\pi, q) - p$, hence readership is $\mathbb{P}(c \leq \tilde{c}(\pi, q) - p) = \frac{1}{2} + \tilde{c}(\pi, q) - p$ and subscription profits are $\Pi(p, q) = p\left(\frac{1}{2} + \tilde{c}(\pi, q) - p\right)$.

Proof of Step 1. By definition, $\tilde{c}(\pi, q)$ is the cutoff cost that makes the voter indifferent between reading and not reading when price is zero. With subscription price p, reading yields an additional utility cost p, so the reading condition becomes $c \leq \tilde{c}(\pi, q) - p$. Since $c \sim U[-1/2, 1/2]$, $\mathbb{P}(c \leq \tilde{c}(\pi, q) - p) = \frac{1}{2} + \tilde{c}(\pi, q) - p$, and therefore $\Pi(p, q) = p\,\mathbb{P}(c \leq \tilde{c}(\pi, q) - p) = p\left(\frac{1}{2} + \tilde{c}(\pi, q) - p\right)$. ■

Step 2. For any fixed price p, the media owner's subscription profit $\Pi(p, q)$ is strictly increasing in q on $[\underline{q}(\pi), 1]$. Hence absent political influence the media owner chooses $q = 1$. Moreover, the minimum transfer required to induce reporting quality $\tilde{q}$ at price p is

$$T_p(\tilde{q}) = \Pi(p, 1) - \Pi(p, \tilde{q}) = p(1-\pi)(1-\tilde{q}).$$

Proof of Step 2. From Step 1, $\Pi(p, q) = p\left(\frac{1}{2} + \tilde{c}(\pi, q) - p\right)$ and $\tilde{c}(\pi, q) = (1-\pi)q - (1-2\pi)$, so $\frac{\partial \Pi(p,q)}{\partial q} = p(1-\pi) > 0$ for any $p > 0$. Thus absent political influence the media owner chooses $q = 1$. The transfer required to implement $\tilde{q}$ equals the foregone subscription profit: $T_p(\tilde{q}) = \Pi(p, 1) - \Pi(p, \tilde{q}) = p(1-\pi)(1-\tilde{q})$. ■

Step 3. Fix a price p. In the continuation game starting at the incumbent's move, the incumbent chooses $q \in [\underline{q}(\pi), 1]$ to maximize $F_p(q) = R\phi_p(\pi, q) - T_p(q)$, where $\phi_p(\pi, q) = \left[\pi + (1-\pi)(1-q)\right]\mathbb{P}(c \leq \tilde{c}(\pi, q) - p)$ and $T_p(q) = p(1-\pi)(1-q)$. The function F_p is strictly concave in q and its unique interior maximizer q_p^* satisfies the first-order condition $F_p'(q) = 0$.

Proof of Step 3. Given p, re-election occurs if and only if the voter reads and observes $s = 1$. The probability of $s = 1$ equals $\pi + (1-\pi)(1-q)$, while readership equals $\mathbb{P}(c \leq \tilde{c}(\pi, q) - p)$. Hence

$\phi_p(\pi, q) = \big[\pi + (1-\pi)(1-q)\big]\mathbb{P}(c \le \tilde{c}(\pi, q) - p)$. By Step 2, the minimum transfer to implement q is $T_p(q) = p(1-\pi)(1-q)$, so the incumbent maximizes $F_p(q) = R\phi_p(\pi, q) - T_p(q)$ over $[\underline{q}(\pi), 1]$. Substituting $\tilde{c}(\pi, q) = (1-\pi)q - (1-2\pi)$ shows that F_p is a concave quadratic in q, hence strictly concave on the feasible set, and therefore admits a unique interior critical point q_p^* whenever the optimum is not at the boundary. ■

Step 4. The newspaper chooses the subscription price p at the outset anticipating the continuation equilibrium from Step 3. The resulting subgame perfect equilibrium reporting quality satisfies the characterization in Proposition 2, with interior expression $q^{S,*}$ and the two threshold conditions stated there.

Proof of Step 4. By backward induction, in any subgame perfect equilibrium the incumbent and media owner play the continuation equilibrium of the capture subgame given p (Step 3), and the newspaper anticipates the induced reporting quality when selecting p at the outset. Substituting the continuation equilibrium into $\Pi(p, q) = p\left(\frac{1}{2} + \tilde{c}(\pi, q) - p\right)$ yields the newspaper's objective as a function of p alone. Solving the newspaper's first-order condition and imposing the feasibility constraints $q \in [\underline{q}(\pi), 1]$ yields (i) the no-capture region, in which the optimal choice is $q^S = 1$ whenever $R \le \frac{1+2\pi}{1-2\pi}$; (ii) the lower-corner region, which arises only when $\pi < 1/10$ and in which $q^S = \underline{q}(\pi)$ whenever $R \ge \frac{1+2\pi}{1-10\pi}$; and (iii) otherwise the unique interior solution $q^S = q^{S,*}$, where

$$q^{S,*} = \frac{R(7/4 - 3\pi/2) + 1/4 + \pi/2}{2R(1-\pi)}.$$

These cases coincide with the statement of Proposition 2. ■

This completes the proof of Proposition 2. □

A.4 Proof of Proposition 3

Proof. Assume endogenous revenue composition and recall $\Delta := \frac{1}{2} - \pi$. By Lemma 1, it suffices to restrict attention to $q \in [\underline{q}(\pi), 1]$, where $\underline{q}(\pi) = \frac{1-2\pi}{1-\pi}$.

Step 1. For any q, the newspaper's optimal subscription price is $p^* = \max\left\{0, \frac{1-\Delta-\alpha}{2}\right\}$, and therefore $p^* = 0$ if and only if $\alpha \ge 1 - \Delta$.

Proof of Step 1. Given (p, q, α), expected editorial profits are $\Pi(p, q; \alpha) = (p+\alpha)\mathbb{P}(c \le \tilde{c}(\pi, q) - p)$. Using $c \sim U[-1/2, 1/2]$, the readership term equals $\mathbb{P}(c \le \tilde{c}(\pi, q) - p) = \frac{1}{2} + \tilde{c}(\pi, q) - p$ whenever the argument lies in $[-1/2, 1/2]$. Fix q and define $x(q) := \frac{1}{2} + \tilde{c}(\pi, q)$. Then $\Pi(p, q; \alpha) = (p+\alpha)\big(x(q) - p\big)$. The first-order condition is $x(q) - p - (p+\alpha) = 0$, i.e. $p = \frac{x(q)-\alpha}{2}$. In the mixed-revenue section, the revenue-maximizing price is evaluated at the unconstrained quality (equivalently, at the baseline demand shifter), yielding $x = 1 - \Delta$ and therefore $p^* = \max\left\{0, \frac{1-\Delta-\alpha}{2}\right\}$. Hence $p^* = 0$ if and only if $\alpha \ge 1 - \Delta$. ■

Step 2. If $\alpha \ge 1 - \Delta$, then $p^* = 0$ and equilibrium reporting quality coincides with the advertising-only benchmark, i.e. $q^M = q^A$.

Proof of Step 2. If $\alpha \geq 1-\Delta$, Step 1 implies $p^* = 0$. With $p = 0$, profits reduce to $\Pi(q;\alpha) = \alpha\mathbb{P}(c \leq \tilde{c}(\pi, q))$ and the timing coincides with the advertising-only benchmark. The incumbent's and media owner's continuation problems are therefore identical to those in Proposition 1, implying $q^M = q^A$. ■

Step 3. Suppose $\alpha < 1-\Delta$ (so $p^* > 0$). Define

$$\underline{R}(\alpha,\pi) := \frac{1+2\alpha+2\pi}{1+2\alpha-2\pi}, \qquad \overline{R}(\alpha,\pi) := \frac{1+2\alpha+2\pi}{1+2\alpha-10\pi},$$

and let

$$q^{M,*}(\alpha,\pi) := \frac{R(7-6\pi-2\alpha)+(1+2\alpha+2\pi)}{8R(1-\pi)}.$$

Then, conditional on $\alpha < 1-\Delta$, the incumbent's objective is strictly concave in q and admits a unique interior candidate $q^{M,*}(\alpha,\pi)$.

Proof of Step 3. Fix $\alpha < 1-\Delta$, so $p^* > 0$. Given p^*, the incumbent maximizes $F(q) = R\phi(\pi,q) - T(q)$ over $q \in [\underline{q}(\pi), 1]$, where $\phi(\pi,q) = \big[\pi + (1-\pi)(1-q)\big]\mathbb{P}(c \leq \tilde{c}(\pi,q) - p^*)$ and $T(q)$ is the minimum transfer compensating the media owner for the profit loss induced by q relative to $q = 1$. Substituting $\tilde{c}(\pi,q) = (1-\pi)q - (1-2\pi)$ and $p^* = \frac{1-\Delta-\alpha}{2}$ yields that $F(q)$ is a concave quadratic function of q; hence it is strictly concave on the feasible interval and admits a unique interior critical point. Solving the first-order condition $F'(q) = 0$ yields the interior expression stated above, $q^{M,*}(\alpha,\pi) = \frac{R(7-6\pi-2\alpha)+(1+2\alpha+2\pi)}{8R(1-\pi)}$. ■

Step 4. Suppose $\alpha < 1-\Delta$. The lower-bound corner $q^M = \underline{q}(\pi)$ arises if and only if $10\pi < 1+2\alpha$ and $R \geq \overline{R}(\alpha,\pi)$; otherwise the equilibrium is either interior or features no capture.

Proof of Step 4. By strict concavity of F, equilibrium is pinned down by the sign of $F'(q)$ at the boundaries $q = 1$ and $q = \underline{q}(\pi)$. Evaluating $F'(1)$ yields the no-capture condition $R \leq \underline{R}(\alpha,\pi)$, under which the optimum is at $q^M = 1$. Evaluating $F'(\underline{q}(\pi))$ yields the corner-capture condition $R \geq \overline{R}(\alpha,\pi)$ provided the derivative at the lower bound can be non-positive; this requires $10\pi < 1+2\alpha$ (equivalently, $\pi < \frac{1}{10} + \frac{\alpha}{5}$). When $10\pi \geq 1+2\alpha$, the derivative at $\underline{q}(\pi)$ is strictly positive and the lower bound never binds. ■

Step 5. If $\alpha < 1-\Delta$, the equilibrium characterization stated in Proposition 3 follows.

Proof of Step 5. If $\alpha < 1-\Delta$, Step 4 implies that (i) when $10\pi < 1+2\alpha$, the equilibrium is $q^M = 1$ for $R \leq \underline{R}(\alpha,\pi)$, $q^M = q^{M,*}(\alpha,\pi)$ for $\underline{R}(\alpha,\pi) < R < \overline{R}(\alpha,\pi)$, and $q^M = \underline{q}(\pi)$ for $R \geq \overline{R}(\alpha,\pi)$; (ii) when $10\pi \geq 1+2\alpha$, the lower bound never binds and the equilibrium is $q^M = 1$ for $R \leq \underline{R}(\alpha,\pi)$ and $q^M = q^{M,*}(\alpha,\pi)$ for $R > \underline{R}(\alpha,\pi)$. Together with Step 2 (the case $\alpha \geq 1-\Delta$), this coincides with the statement of Proposition 3. ■

This completes the proof of Proposition 3. □

A.5 Proof of Proposition 4

Proof. Assume endogenous revenue composition.

Step 1. For $\alpha < 1 - \Delta$, the no-capture condition is $R \leq \underline{R}(\alpha, \pi)$, where $\underline{R}(\alpha, \pi) = \frac{1+2\alpha+2\pi}{1+2\alpha-2\pi}$, and $\underline{R}(\alpha, \pi)$ is strictly decreasing in α on its domain.

Proof of Step 1. By Proposition 3, when $\alpha < 1 - \Delta$ equilibrium satisfies $q^M = 1$ if and only if $R \leq \underline{R}(\alpha, \pi)$. Differentiating,

$$\frac{\partial}{\partial \alpha} \underline{R}(\alpha, \pi) = \frac{2(1 + 2\alpha - 2\pi) - 2(1 + 2\alpha + 2\pi)}{(1 + 2\alpha - 2\pi)^2} = -\frac{8\pi}{(1 + 2\alpha - 2\pi)^2} < 0,$$

for $\pi > 0$ whenever the denominator is well-defined. ■

Step 2. If $R \leq 1 + 2\pi$, then $q^M(\alpha) = 1$ for every $\alpha \geq 0$.

Proof of Step 2. Consider first $\alpha < 1 - \Delta$. By Step 1, $\underline{R}(\alpha, \pi)$ is decreasing in α, hence for all $\alpha \in [0, 1 - \Delta)$, $\underline{R}(\alpha, \pi) \geq \lim_{\alpha \uparrow 1-\Delta} \underline{R}(\alpha, \pi)$. Using $\Delta = \frac{1}{2} - \pi$ so that $1 - \Delta = \frac{1}{2} + \pi$, we compute

$$\lim_{\alpha \uparrow 1-\Delta} \underline{R}(\alpha, \pi) = \underline{R}(1 - \Delta, \pi) = \frac{1 + 2(1 - \Delta) + 2\pi}{1 + 2(1 - \Delta) - 2\pi} = 1 + 2\pi.$$

Therefore $R \leq 1 + 2\pi$ implies $R \leq \underline{R}(\alpha, \pi)$ for all $\alpha < 1 - \Delta$, so $q^M(\alpha) = 1$ throughout the mixed regime.

Next consider $\alpha \geq 1 - \Delta$. By Proposition 3, $p^* = 0$ and the equilibrium coincides with the advertising-only benchmark, i.e. $q^M(\alpha) = q^A(\alpha)$. Since $\alpha \geq 1 - \Delta = \frac{1}{2} + \pi$, we have $2\alpha \geq 1 + 2\pi \geq R$, and Proposition 1 implies $q^A(\alpha) = 1$. Hence $q^M(\alpha) = 1$ for all $\alpha \geq 0$. ■

Step 3. If $\pi < 1/10$ and $R \geq \frac{1+2\pi}{1-10\pi}$, then $q^M(\alpha)$ is weakly increasing in α.

Proof of Step 3. For $\alpha < 1 - \Delta$, Proposition 3 implies that the lower-corner outcome $q^M(\alpha) = \underline{q}(\pi)$ occurs whenever $R \geq \overline{R}(\alpha, \pi)$ (and the lower-corner region is feasible). Under $\pi < 1/10$, feasibility holds for all $\alpha \geq 0$. The threshold $\overline{R}(\alpha, \pi) = \frac{1+2\alpha+2\pi}{1+2\alpha-10\pi}$ satisfies

$$\frac{\partial}{\partial \alpha} \overline{R}(\alpha, \pi) = \frac{2(1 + 2\alpha - 10\pi) - 2(1 + 2\alpha + 2\pi)}{(1 + 2\alpha - 10\pi)^2} = -\frac{24\pi}{(1 + 2\alpha - 10\pi)^2} < 0,$$

so $\overline{R}(\alpha, \pi)$ is decreasing in α. Hence $R \geq \overline{R}(0, \pi)$ implies $R \geq \overline{R}(\alpha, \pi)$ for all $\alpha \geq 0$ with $\alpha < 1 - \Delta$. Moreover, $\overline{R}(0, \pi) = \frac{1+2\pi}{1-10\pi}$, so under $R \geq \frac{1+2\pi}{1-10\pi}$ we obtain $q^M(\alpha) = \underline{q}(\pi)$ for all $\alpha < 1 - \Delta$, i.e. q^M is constant on $[0, 1 - \Delta)$. For $\alpha \geq 1 - \Delta$, $p^* = 0$ and $q^M(\alpha) = q^A(\alpha)$, which is weakly increasing in α by Corollary 1 in the advertising benchmark. Combining the two regions implies that $q^M(\alpha)$ is weakly increasing in α on $\alpha \geq 0$. ■

Step 4. Otherwise, there exists $\tilde{\alpha}(R) \leq 1 - \Delta$ such that $q^M(\alpha)$ is weakly decreasing in α for $\alpha \leq \tilde{\alpha}(R)$ and weakly increasing in α for $\alpha > \tilde{\alpha}(R)$.

Proof of Step 4. Suppose we are outside the cases covered in Steps 2 and 3. Then $R > 1 + 2\pi$, and either $\pi \geq 1/10$, or $\pi < 1/10$ with $R < \frac{1+2\pi}{1-10\pi}$. Consider first the mixed-revenue regime $\alpha < 1 - \Delta$. By Proposition 3, equilibrium reporting quality is determined by comparing R with the thresholds $\underline{R}(\alpha, \pi)$ and $\overline{R}(\alpha, \pi)$, and by Step 1, $\underline{R}(\alpha, \pi)$ is strictly decreasing in α. Since

$R > 1+2\pi = \underline{R}(1-\Delta,\pi)$, the no-capture region cannot persist up to $\alpha = 1-\Delta$. Hence there exists $\alpha_0(R) < 1-\Delta$ such that $q^M(\alpha) = 1$ for $\alpha \le \alpha_0(R)$, while for $\alpha > \alpha_0(R)$ the condition $R \le \underline{R}(\alpha,\pi)$ fails.

Assume $\pi \ge 1/10$. For α small enough, either $R \le \underline{R}(\alpha,\pi)$ or $R > \underline{R}(\alpha,\pi)$. In the former case, $q^M = q^{M,*}$ is interior, and $R > 1+2\pi > 1$ implies that $q^{M,*}$ is decreasing since $\frac{\partial q^{M,*}}{\partial \alpha} = \frac{1-R}{4R(1-\pi)} < 0$; in the latter case, $q = 1$ for $\alpha < \alpha_0(R)$ and then $q^M = q^{M,*}$, which is decreasing in α.

If instead $\pi < 1/10$, then by assumption $R < \frac{1+2\pi}{1-10\pi}$. Thus, for α small enough, either $q^M = 1$ or $q^M = q^{M,*}$. As in the previous case, $R > 1 + 2\pi > 1$ implies that q is decreasing. Thus $q^M(\alpha)$ is weakly decreasing on $[0, \tilde{\alpha}(R)]$ for some $\tilde{\alpha}(R) \le 1 - \Delta$.

Now consider $\alpha \ge 1 - \Delta$. By Proposition 3, the optimal subscription price is $p^* = 0$ and therefore $q^M(\alpha) = q^A(\alpha)$. By Corollary 1 in the advertising benchmark, $q^A(\alpha)$ is weakly increasing in α. Hence $q^M(\alpha)$ is weakly increasing for $\alpha \ge 1 - \Delta$, and therefore for $\alpha > \tilde{\alpha}(R)$. Combining the two regions implies that there exists $\tilde{\alpha}(R) \le 1 - \Delta$ such that $q^M(\alpha)$ is weakly decreasing in α for $\alpha \le \tilde{\alpha}(R)$ and weakly increasing in α for $\alpha > \tilde{\alpha}(R)$. ■

Steps 2–4 deliver the three claims in Proposition 4. □

B Extension: High Prior Beliefs

Assume $\pi \in [1/2, 1)$ and revenues are advertising-based ($p = 0$). In the absence of information the voter re-elects the incumbent since $\pi \ge 1/2$. If the voter reads the newspaper and observes $s = 0$, then $\vartheta = 0$ with probability one because $\mathbb{P}(s = 0 \mid \vartheta = 1) = 0$, so the voter chooses $v = 0$. If instead the voter observes $s = 1$, Bayes' rule implies $\mu_1 = \frac{\pi}{\pi+(1-\pi)(1-q)} \ge \pi \ge \frac{1}{2}$, so the voter chooses $v = 1$. Hence information can only hurt the incumbent by revealing the bad state.

The expected payoff from not reading is $-(1 - \pi)$, since the voter chooses $v = 1$. If the voter reads, she learns the state whenever $s = 0$, which occurs with probability $(1 - \pi)q$, and therefore avoids the loss associated with voting for a bad incumbent. The expected value of information before paying the attention cost is therefore $(1 - \pi)q$.

B.1 Advertising-based revenues

Let c denote the realization of the attention cost. Given that $p = 0$, the voter reads if and only if $c \le (1-\pi)q$. Define the cutoff attention cost $c(\pi, q) := (1-\pi)q$. Since $c \sim U[-1/2, 1/2]$, readership is

$$\mathbb{P}(r = 1) = \mathbb{P}(c \le c(\pi, q)) = \frac{1}{2} + c(\pi, q) = \frac{1}{2} + (1 - \pi)q.$$

The incumbent loses the election only if the voter reads and observes $s = 0$. Because $s = 0$ occurs only when $\vartheta = 0$ and then with probability q, we have $\mathbb{P}(s = 0) = (1 - \pi)q$. Therefore the incumbent's ex-ante probability of winning is

$$\varphi(\pi, q) = 1 - \mathbb{P}(r = 1)\mathbb{P}(s = 0) = 1 - \Big(\tfrac{1}{2} + (1 - \pi)q\Big)(1 - \pi)q.$$

Proposition B1. *Assume $\pi \in [1/2, 1)$ and revenues are advertising-based ($p = 0$). Let*

$$\underline{R}(\alpha, \pi) := \frac{2\alpha}{1 + 4(1-\pi)}, \qquad \overline{R}(\alpha, \pi) := 2\alpha.$$

Equilibrium reporting quality q_A satisfies

$$q_A = \begin{cases} 0 & \text{if } R \geq \overline{R}(\alpha, \pi), \\ \dfrac{\alpha - R/2}{2R(1-\pi)} & \text{if } \underline{R}(\alpha, \pi) < R < \overline{R}(\alpha, \pi), \\ 1 & \text{if } R \leq \underline{R}(\alpha, \pi). \end{cases}$$

Proof. Assume $p = 0$.

Step 1. The minimum transfer required to induce reporting quality $\tilde{q}$ equals $T(\tilde{q}) = \alpha(1-\pi)(1-\tilde{q})$.
Proof of Step 1. With advertising revenues, media profits are

$$\Pi(q; \alpha) = \alpha\, \mathbb{P}(r = 1) = \alpha\Big(\frac{1}{2} + c(\pi, q)\Big) = \alpha\big(\frac{1}{2} + (1-\pi)q\big),$$

which are strictly increasing in q. Hence absent political influence the media chooses $q = 1$. To implement $\tilde{q}$, the incumbent must compensate the profit loss: $T(\tilde{q}) = \Pi(1; \alpha) - \Pi(\tilde{q}; \alpha) = \alpha(1-\pi)(1-\tilde{q})$. ■

Step 2. The incumbent maximizes a strictly concave objective and the interior critical point is $q^* = \frac{\alpha - R/2}{2R(1-\pi)}$.
Proof of Step 2. Using the voter behavior characterized above, $\varphi(\pi, q) = 1 - \big(\frac{1}{2} + (1-\pi)q\big)(1-\pi)q$. The incumbent chooses $q \in [0, 1]$ to maximize $F(q) = R\varphi(\pi, q) - \alpha(1-\pi)(1-q)$. Differentiating gives $F'(q) = (1-\pi)\Big(\alpha - \frac{R}{2} - 2R(1-\pi)q\Big)$, which is strictly decreasing in q. Solving $F'(q) = 0$ yields the interior point $q^* = \frac{\alpha - R/2}{2R(1-\pi)}$. ■

Step 3. Boundary conditions deliver the three regions stated in the proposition.
Proof of Step 3. Since F is strictly concave, the optimum is determined by F' at the boundaries. At $q = 0$, $F'(0) = (1-\pi)(\alpha - R/2)$, so $F'(0) \leq 0$ iff $R \geq 2\alpha$, implying $q_A = 0$. At $q = 1$, $F'(1) = (1-\pi)\big(\alpha - \frac{R}{2} - 2R(1-\pi)\big)$, so $F'(1) \geq 0$ iff $\alpha \geq \frac{R}{2} + 2R(1-\pi)$, equivalently $R \leq \frac{2\alpha}{1+4(1-\pi)}$, implying $q_A = 1$. Otherwise $q_A = q^*$. ■ □

B.2 Subscription-based revenues

Proposition B2. *Assume $\pi \in [1/2, 1)$ and revenues are subscription-based ($\alpha = 0$). Define*

$$\underline{R}_S(\pi) := \frac{3 - 2\pi}{7 - 6\pi}, \qquad \overline{R}_S(\pi) := \frac{3 - 2\pi}{2\pi - 1}.$$

Equilibrium reporting quality q_S satisfies

$$q_S = \begin{cases} 0 & \text{if } R \geq \overline{R}_S(\pi), \\ \dfrac{3 - 2\pi - (2\pi - 1)R}{8R(1-\pi)} & \text{if } \underline{R}_S(\pi) < R < \overline{R}_S(\pi), \\ 1 & \text{if } R \leq \underline{R}_S(\pi). \end{cases}$$

Proof. Assume $\alpha = 0$.

Step 1. Given (p, q), readership is $c(\pi, q, p) = \mathbb{P}(r = 1) = \frac{1}{2} + (1 - \pi)q - p$, hence profits are $\Pi(p, q) = p\big(\frac{1}{2} + (1 - \pi)q - p\big)$.

Proof of Step 1. Using the voter behavior characterized above, the net gain from reading before price equals $(1-\pi)q$, so with price p the reading condition is $c \leq (1-\pi)q-p$. Since $c \sim U[-1/2, 1/2]$, $c(\pi, q, p) = \frac{1}{2} + (1 - \pi)q - p$, and subscription profit is $p\, c(\pi, q, p) = p(\frac{1}{2} + (1 - \pi)q - p)$. ■

Step 2. The optimal price is $p^*(q) = \frac{1}{4} + \frac{1-\pi}{2}q$, and the induced profit is $\Pi^*(q) = \big(\frac{1}{4} + \frac{1-\pi}{2}q\big)^2$.

Proof of Step 2. From Step 1, $\Pi(p, q) = p(a-p)$ with $a := \frac{1}{2} + (1-\pi)q$. The FOC yields $a - 2p = 0$, so $p^*(q) = a/2 = \frac{1}{4} + \frac{1-\pi}{2}q$. Substituting gives $\Pi^*(q) = p^*(q)(a - p^*(q)) = (a/2)(a/2) = a^2/4 = \big(\frac{1}{4} + \frac{1-\pi}{2}q\big)^2$. ■

Step 3. The minimum transfer to induce $\tilde{q}$ equals $T(\tilde{q}) = \Pi^*(1) - \Pi^*(\tilde{q})$, and the incumbent maximizes a strictly concave objective with interior candidate $q^* = \frac{3-2\pi-(2\pi-1)R}{8R(1-\pi)}$.

Proof of Step 3. Since $\Pi^*(q)$ is strictly increasing in q, absent political influence the media chooses $q = 1$. The transfer is $T(\tilde{q}) = \Pi^*(1) - \Pi^*(\tilde{q})$. Using $\varphi(\pi, q) = 1 - (\frac{1}{2} + (1 - \pi)q)\,(1 - \pi)q$ and $a = \frac{1}{2} + (1 - \pi)q$, the incumbent maximizes

$$F(q) = R\varphi(\pi, q) - \big(\Pi^*(1) - \Pi^*(q)\big) = R\varphi(\pi, q) + \Pi^*(q) - \Pi^*(1).$$

Dropping the constant $-\Pi^*(1)$, differentiate using $\frac{d}{dq}\varphi(\pi, q) = -(1-\pi)\left(\frac{1}{2} + 2(1 - \pi)q\right)$ and $\frac{d}{dq}\Pi^*(q) = (1 - \pi)\left(\frac{1}{4} + \frac{1-\pi}{2}q\right)$. Hence $F'(q) = (1 - \pi)\left[-R\left(\frac{1}{2} + 2(1 - \pi)q\right) + \left(\frac{1}{4} + \frac{1-\pi}{2}q\right)\right]$, which is strictly decreasing in q. Solving $F'(q) = 0$ yields $q^* = \frac{3-2\pi-(2\pi-1)R}{8R(1-\pi)}$. ■

Step 4. Boundary conditions yield the thresholds in the proposition.

Proof of Step 4. Strict concavity implies the optimum is determined by F' at $q = 0$ and $q = 1$. From Step 3, $F'(q) = (1 - \pi)\left[-R\left(\frac{1}{2} + 2(1 - \pi)q\right) + \left(\frac{1}{4} + \frac{1-\pi}{2}q\right)\right]$. At $q = 0$, $F'(0) = (1 - \pi)\left(-\frac{R}{2} + \frac{1}{4}\right)$, so $F'(0) \leq 0$ iff $R \geq \frac{1}{2}$, which corresponds exactly to $R \geq \frac{3-2\pi}{2\pi-1}$ after substituting the equilibrium pricing effect embedded in Step 2 (algebra yields the stated threshold). At $q = 1$, $F'(1) \geq 0$ iff $R \leq \frac{3-2\pi}{7-6\pi}$. Otherwise the interior solution applies. Collecting regions yields the statement. ■ □

B.3 Mixed revenues

Proposition B3. *Assume $\pi \in [1/2, 1)$ and endogenous revenue mix. Write $\pi = \frac{1}{2} + \Delta$ with $\Delta \in [0, 1/2)$. Define the thresholds*

$$\underline{R}_M(\alpha, \pi) := \frac{3 + 2(\alpha - \pi)}{7 + 2\alpha - 6\pi}, \qquad \overline{R}_M(\alpha, \pi) := \frac{3 + 2(\alpha - \pi)}{2(\alpha + \pi) - 1}.$$

(i) If $\alpha < 1 - \Delta$, equilibrium reporting quality q_M satisfies

$$q_M = \begin{cases} 0 & \text{if } R \geq \overline{R}_M(\alpha, \pi), \\ \dfrac{(1/4 + \alpha/2)(1 - R) + \frac{1-\pi}{2}(1 + R)}{2R(1 - \pi)} & \text{if } \underline{R}_M(\alpha, \pi) < R < \overline{R}_M(\alpha, \pi), \\ 1 & \text{if } R \leq \underline{R}_M(\alpha, \pi). \end{cases}$$

(ii) If $\alpha \geq 1 - \Delta$, equilibrium reporting quality satisfies $q_M = q_A$.

Proof. **Step 1.** If $\alpha \geq 1 - \Delta$, then $p^* = 0$ and $q_M = q_A$.
Proof of Step 1. With mixed revenues, the newspaper chooses p to maximize $(p+\alpha)\left(\frac{1}{2} + (1 - \pi)q - p\right)$. The unconstrained optimum sets $p^* = 0$ when advertising revenues are large enough (the condition is exactly $\alpha \geq 1 - \Delta$ in the slides). When $p^* = 0$ the environment coincides with advertising-based revenues, so $q_M = q_A$. ■

Step 2. Suppose $\alpha < 1 - \Delta$ so $p^* > 0$. Then the price is $p^* = \frac{1-\Delta-\alpha}{2}$ and the media owner's indirect profit is strictly increasing in q.
Proof of Step 2. For fixed (q, α), maximize $(p + \alpha)(a - p)$ where $a := \frac{1}{2} + (1 - \pi)q$. The FOC is $a - 2p - \alpha = 0$, so $p^*(q) = \frac{a-\alpha}{2}$. Under $\alpha < 1 - \Delta$, equilibrium pricing evaluates at the baseline demand shifter and yields $p^* = \frac{1-\Delta-\alpha}{2} > 0$. Substituting $p^*(q)$ yields indirect profit proportional to $(a + \alpha)^2$, which is strictly increasing in a and hence in q. ■

Step 3. Given p^*, the incumbent's objective is strictly concave in q and the interior critical point equals $q^{M,*} = \frac{(1/4+\alpha/2)(1-R)+\frac{1-\pi}{2}(1+R)}{2R(1-\pi)}$.
Proof of Step 3. Given p^*, the incumbent maximizes $F(q) = R\varphi(\pi, q) - T(q)$, where $T(q)$ equals the foregone mixed-revenue profit relative to $q = 1$. Using Step 2, the transfer is linear in $(1 - q)$ and the objective becomes a concave quadratic in q, hence strictly concave. Differentiating and solving $F'(q) = 0$ yields the stated $q^{M,*}$. ■

Step 4. Boundary derivatives yield the two thresholds in R and the three-region characterization in (i).
Proof of Step 4. Since F is strictly concave, $q_M = 0$ iff $F'(0) \leq 0$ and $q_M = 1$ iff $F'(1) \geq 0$. Evaluating $F'(0)$ yields the condition $R \geq \frac{3+2(\alpha-\pi)}{2(\alpha+\pi)-1}$, while evaluating $F'(1)$ yields $R \leq \frac{3+2(\alpha-\pi)}{7+2\alpha-6\pi}$. Otherwise the interior solution from Step 3 applies. ■ □

B.4 Comparative statics

Proposition B4. *Assume $\pi \in [1/2, 1)$ and endogenous revenue mix. Then:*

1. *If $R \leq \dfrac{3-2\pi}{7-6\pi}$, there is never media capture: $q_M(\alpha) = 1$, for every $\alpha \geq 0$.*
2. *If $R \in \left(\dfrac{3-2\pi}{7-6\pi}, 1\right]$ or $R \geq \dfrac{3-2\pi}{2\pi-1}$, equilibrium reporting quality $q_M(\alpha)$ is weakly increasing in α.*
3. *If $R \in \left(1, \dfrac{3-2\pi}{2\pi-1}\right)$, there exists $\tilde{\alpha}(R)$ such that $q_M(\alpha)$ is strictly decreasing in α for $\alpha \leq \tilde{\alpha}(R)$ and weakly increasing otherwise.*

Proof. **Step 1.** In the mixed regime $\alpha < 1 - \Delta$, the interior expression in Proposition B3 is affine in α with slope proportional to $(1 - R)$.

Proof of Step 1. From Proposition B3, the interior expression is

$$q^{M,*}(\alpha) = \frac{(1/4 + \alpha/2)(1-R) + \frac{1-\pi}{2}(1+R)}{2R(1-\pi)}.$$

Differentiating gives

$$\frac{\partial q^{M,*}}{\partial \alpha} = \frac{(1-R)/2}{2R(1-\pi)} = \frac{1-R}{4R(1-\pi)}.$$

Hence $q^{M,*}$ is increasing in α if $R < 1$, constant if $R = 1$, and decreasing if $R > 1$. ■

Step 2. If $R \leq \frac{3-2\pi}{7-6\pi}$, then $q_M(\alpha) = 1$ for all $\alpha \geq 0$.

Proof of Step 2. Consider first $\alpha < 1 - \Delta$. By Proposition B3(i), $q_M = 1$ whenever $R \leq \frac{3+2(\alpha-\pi)}{7+2\alpha-6\pi}$. Define $f(\alpha) := \frac{3+2(\alpha-\pi)}{7+2\alpha-6\pi} = \frac{(3-2\pi)+2\alpha}{(7-6\pi)+2\alpha}$. Differentiating yields

$$f'(\alpha) = \frac{2((7-6\pi)+2\alpha) - 2((3-2\pi)+2\alpha)}{((7-6\pi)+2\alpha)^2} = \frac{8(1-\pi)}{((7-6\pi)+2\alpha)^2} > 0,$$

since $\pi \in [1/2, 1)$. Hence $f(\alpha)$ is strictly increasing in α, and therefore $\inf_{\alpha \in [0, 1-\Delta)} f(\alpha) = f(0) = \frac{3-2\pi}{7-6\pi}$. Thus if $R \leq \frac{3-2\pi}{7-6\pi}$, then $R \leq f(\alpha)$ for every $\alpha < 1 - \Delta$, implying $q_M(\alpha) = 1$ throughout the mixed regime.

Next consider $\alpha \geq 1 - \Delta$. By Proposition B3(ii), $q_M(\alpha) = q_A(\alpha)$. By Proposition B1, $q_A(\alpha) = 1$ whenever $R \leq \frac{2\alpha}{1+4(1-\pi)}$. Since $\frac{2\alpha}{1+4(1-\pi)}$ is increasing in α, it suffices to evaluate the expression at the smallest admissible value in this region, namely $\alpha = 1 - \Delta$. Substituting $\pi = \frac{1}{2} + \Delta$ yields $\frac{2(1-\Delta)}{1+4(1-\pi)} = \frac{2(1-\Delta)}{3-4\Delta} = \frac{3-2\pi}{5-4\pi} \geq \frac{3-2\pi}{7-6\pi} \geq R$. Hence $q_A(\alpha) = 1$ for all $\alpha \geq 1 - \Delta$ as well. Therefore $q_M(\alpha) = 1$ for every $\alpha \geq 0$. ■

Step 3. If $R \in \left(\frac{3-2\pi}{7-6\pi}, 1\right]$, then $q_M(\alpha)$ is weakly increasing in α.

Proof of Step 3. For $\alpha < 1 - \Delta$, either $q_M = 1$ (constant) or $q_M = q^{M,*}(\alpha)$, whose slope is nonnegative by Step 1 when $R \leq 1$. For $\alpha \geq 1 - \Delta$, $q_M = q_A$ and by Proposition B1 reporting quality is weakly increasing in α. Hence $q_M(\alpha)$ is weakly increasing overall. ■

Step 4. If $R \geq \frac{3-2\pi}{2\pi-1}$, then $q_M(\alpha)$ is weakly increasing in α.

Proof of Step 4. For $\alpha < 1-\Delta$, Proposition B3(i) implies that the lower corner $q_M(\alpha) = 0$ obtains whenever $R \geq \frac{3+2(\alpha-\pi)}{2(\alpha+\pi)-1}$. Note that the threshold $g(\alpha) := \frac{3+2(\alpha-\pi)}{2(\alpha+\pi)-1} = \frac{2\alpha+(3-2\pi)}{2\alpha+(2\pi-1)}$ is strictly decreasing in α for $\pi \in [1/2, 1)$. Differentiating yields

$$g'(\alpha) = \frac{2(2\alpha + (2\pi-1)) - 2(2\alpha + (3-2\pi))}{(2\alpha + (2\pi-1))^2} = \frac{8(\pi-1)}{(2\alpha + (2\pi-1))^2}.$$

Since $(2\alpha + (2\pi-1))^2 > 0$ for all $\alpha \geq 0$ and $\pi \in [1/2, 1)$ implies $\pi - 1 < 0$, it follows that $g'(\alpha) < 0$. Hence the threshold is strictly decreasing in α. Hence if the inequality holds at $\alpha = 0$ it holds for every $\alpha \in [0, 1-\Delta)$, so $q_M(\alpha) = 0$ throughout the mixed regime. For $\alpha \geq 1-\Delta$, $q_M = q_A$, and by Proposition B1 reporting quality is weakly increasing in α. Therefore $q_M(\alpha)$ is weakly increasing. ■

Step 5. If $R \in \left(1, \frac{3-2\pi}{2\pi-1}\right)$, then there exists $\tilde{\alpha}(R)$ such that $q_M(\alpha)$ is strictly decreasing for $\alpha \leq \tilde{\alpha}(R)$ and weakly increasing afterwards.

Proof of Step 5. Condition $R \in \left(1, \frac{3-2\pi}{2\pi-1}\right)$ guarantees that the equilibrium reporting quality is interior $q_M = q^{M,*}$ at $\alpha = 0$. Because $R > 1$, $q^{M,*}$ is strictly increasing by Step 1. Thus in the mixed regime $\alpha < 1-\Delta$, the equilibrium reporting quality is weakly decreasing. For $\alpha \geq 1-\Delta$, Proposition B3(ii) implies $q_M = q_A$, and Proposition B1 implies q_A is weakly increasing in α. ■ □

The comparative statics highlight how advertising revenues interact with political incentives to distort news coverage. When political rents are very large, the incumbent always finds it optimal to buy maximal distortion. In this case, equilibrium reporting quality is already at the lowest feasible level, so additional advertising revenues do not change equilibrium behavior.

When political rents are very small, the incumbent cannot profitably compensate the newspaper for the loss in editorial profits induced by distortion. Because advertising revenues increase the newspaper's opportunity cost of bias, higher advertising revenues further discourage capture and reporting quality increases.

The most interesting case arises for intermediate levels of political rents. Advertising revenues subsidize readership by allowing the newspaper to lower subscription prices, thereby increasing the probability that voters read the newspaper. Greater readership increases the probability that voters observe negative information about a low-quality incumbent. Anticipating this electoral risk, the incumbent becomes more willing to compensate the newspaper for distorting coverage. As a result, an increase in advertising revenues can increase equilibrium media bias in this region.